Assignment 2

Q1. Data-link layer responsibilities; LLC vs MAC

The data-link layer provides reliable node-to-node (hop-to-hop) delivery over a single link or LAN.

Responsibilities

- **Framing:** encapsulates network-layer packets into frames with a header and trailer.
- **Physical addressing:** uses MAC addresses to identify sender and receiver on the same link.
- **Error detection:** a CRC/FCS in the trailer detects corrupted frames; some protocols also correct or retransmit.
- **Flow control:** stops a fast sender from overwhelming a slow receiver.
- **Media access control:** decides who may transmit on a shared medium (CSMA/CD, CSMA/CA).

LLC vs MAC (IEEE 802 sublayers)

Aspect	LLC (802.2)	MAC (802.3 / 802.11)
Position	Upper sublayer	Lower sublayer
Role	Interface to the network layer; multiplexes protocols (SAP/SNAP); optional flow and error control	Framing, MAC addressing, medium access, FCS
Dependence	Independent of the medium	Specific to the technology (Ethernet, Wi-Fi)
Talks to	Network layer above	Physical layer below

Q2. Ethernet frame format

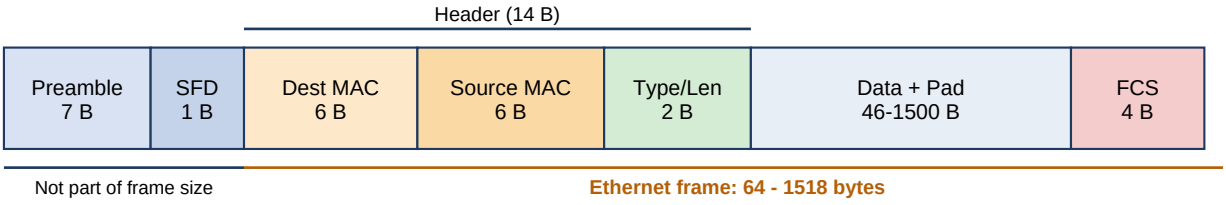

Figure 1: Ethernet (802.3) frame format

Field	Size	Purpose
Preamble	7 B	Alternating 1s and 0s (10101010) for receiver clock synchronization
SFD	1 B	Start Frame Delimiter (10101011); marks the start of the frame
Destination MAC	6 B	Receiver's hardware address: unicast, multicast, or broadcast (FF:FF:FF:FF:FF:FF)
Source MAC	6 B	Sender's hardware address; always unicast
Type/Length	2 B	Value >= 1536 (0x0600) names the upper-layer protocol, e.g. 0x0800 IPv4, 0x0806 ARP, 0x86DD IPv6. Smaller values give the payload length (802.3)
Data/Payload	46-1500 B	Network-layer packet; padded if under 46 B
FCS	4 B	CRC-32 over header and data for error detection; bad frames are dropped

Minimum frame size is 64 B and maximum is 1518 B (excluding preamble and SFD).

Q4. Switch operation and MAC learning

A switch is a layer-2 device that forwards frames using a **MAC address table (forwarding table)**, which it builds dynamically:

- **Learn:** for every incoming frame, record *source MAC to incoming port*.
- **Forward:** look up the *destination MAC*. If found, send out that port only (filtering). If unknown unicast, broadcast or multicast, **flood** out all ports except the incoming one.
- **Filter:** if the destination is on the same port as the source, drop the frame.
- **Age:** entries expire after about 300 s without traffic.

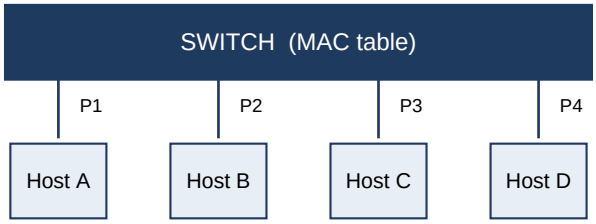

Figure 2: Hosts A-D on ports P1-P4; the table starts empty

Step	Frame	Switch action	Table after
1	A to B	Learns A on P1. B unknown, so floods P2, P3, P4	A-P1
2	B to A	Learns B on P2. A known, forwards only to P1	A-P1, B-P2
3	C to A	Learns C on P3. Forwards to P1 only	A-P1, B-P2, C-P3
4	D to C	Learns D on P4. Forwards to P3 only	A-P1, B-P2, C-P3, D-P4

On a Cisco switch, `show mac address-table` displays this table.

Q5. Hub vs switch vs bridge

Feature	Hub	Bridge	Switch
OSI layer	1 (Physical)	2 (Data-link)	2 (Data-link); some are L3
Forwarding	Repeats bits to all ports	By MAC table, software based	By MAC table, hardware based (ASIC)
Collision domains	One for all ports	One per port/segment	One per port
Broadcast domains	1	1	1 per VLAN
Duplex	Half only	Mostly half	Full duplex
Ports	Few	2-4	Many (8-48+)
Efficiency	Low, shared bandwidth	Moderate	High, dedicated bandwidth
Practical use	Obsolete	Joining two LAN segments; legacy	Standard in all modern LANs

Q6. CSMA/CD and why it matters less today

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) is used in classic half-duplex shared Ethernet:

- **Carrier sense:** listen before sending; transmit if the medium is idle.
- **Multiple access:** many stations share the same medium.
- **Collision detection:** while sending, monitor the line; a differing signal means a collision.
- **Jam signal:** sent so all stations notice the collision.
- **Binary exponential backoff:** after the n-th collision wait a random 0 to (2^n - 1) slot times (n capped at 10); give up after 16 attempts.

The 64 B minimum frame size ensures the sender is still transmitting when a collision signal returns.

Why it is less important now

- Switches give every port its own **collision domain**, so there is no shared medium.
- **Full-duplex** links use separate transmit and receive pairs, so stations send and receive simultaneously without collisions.
- Gigabit and faster Ethernet is effectively always full duplex; CSMA/CD is not used and remains in the standard only for half-duplex compatibility.

Q7. CSMA/CA in IEEE 802.11 and comparison with CSMA/CD

Wireless stations **cannot detect collisions** while transmitting, because their own signal drowns out others, and the **hidden-node problem** exists. So 802.11 **avoids** collisions instead of detecting them.

Operation (DCF)

1. Sense the channel. If idle for **DIFS**, begin the backoff procedure.
2. If busy, wait until idle, then pick a random backoff counter that decrements only while the channel is idle.
3. When the counter reaches 0, transmit the frame.
4. The receiver replies with an **ACK** after **SIFS**. No ACK means collision or loss, so the sender doubles its contention window and retries.
5. **Optional RTS/CTS:** sender sends RTS, receiver answers CTS, and other stations set their **NAV** (virtual carrier sense) and stay silent. This solves hidden nodes.

Aspect	CSMA/CD	CSMA/CA
Objective	Detect collisions and recover	Avoid collisions beforehand
Medium	Wired (half-duplex Ethernet)	Wireless (802.11)
Detection	Monitors the line during transmission	Cannot detect; relies on ACKs
Acknowledgement	None at the MAC layer	Mandatory ACK per frame
Mechanisms	Jam signal, backoff after collision	Backoff before sending, IFS, NAV, RTS/CTS
Overhead	Low	Higher

Q8. Wireless LAN design for a college laboratory

Assumed scenario: one lab of about 12 m x 10 m with 40 client devices (laptops and phones).

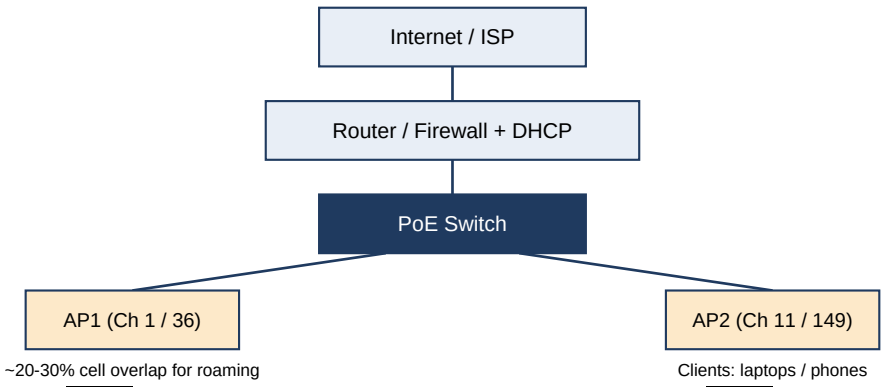

Figure 3: Proposed WLAN topology

Aspect	Specification
AP count and placement	2 dual-band Wi-Fi 6 APs, **ceiling-mounted** at about 2.5-3 m, near the centre of each half of the lab. Keep away from metal cabinets and thick walls. Powered via PoE from the switch with Ethernet back-haul.
SSID	`CollegeLab-WiFi` (staff/students, WPA2/WPA3-Enterprise) and `CollegeLab-Guest` (isolated guest network). The same SSID on both APs lets clients roam.
Channels	2.4 GHz: non-overlapping channels **1, 6, 11** (AP1 = 1, AP2 = 11). 5 GHz: channels 36 and 149 (20/40 MHz width) to reduce interference; prefer 5 GHz for clients. Adjacent APs use different channels with about 20-30% cell overlap for roaming. Run a site survey (Wi-Fi analyzer) before finalizing.
Addressing	Private subnet **192.168.10.0/24** (254 hosts). Gateway 192.168.10.1; DHCP pool 192.168.10.50-.250; static addresses for APs/printers in .2-.20. Guest: 192.168.20.0/24 on a separate VLAN. DNS via the college server or 8.8.8.8.
Security	WPA3 (or WPA2-AES minimum); 802.1X with RADIUS authentication; no WEP/TKIP; disable WPS; strong passphrase if PSK is used; change default AP admin password; MAC filtering only as a supplement; guest VLAN isolation and client isolation; firewall rules; regular firmware updates; logging and monitoring.

Q9. Mobile IP and the need for mobility support

Problem: an IP address encodes the host's network location. When a device moves to another network it needs a new IP address, which **breaks ongoing TCP connections** and makes the device unreachable at its old address. Mobile IP lets a device keep a **permanent address** while moving.

Components

- **Mobile Node (MN):** the moving device, with a permanent *home address*.
- **Home Agent (HA):** router on the home network that tracks the MN's location and tunnels packets to it.
- **Foreign Agent (FA):** router on the visited network that provides a care-of address.
- **Care-of Address (CoA):** temporary address of the MN on the foreign network (FA-provided or co-located).

Operation

1. **Agent discovery:** the MN learns it is away from home through agent advertisements (ICMP router discovery).
2. **Registration:** the MN registers its CoA with the HA, directly or through the FA.
3. **Tunneling:** the HA intercepts packets for the home address and **encapsulates** them (IP-in-IP) to the CoA; the FA or MN de-encapsulates and delivers them.
4. **Reverse path:** the MN replies directly to the correspondent (triangle routing), or via reverse tunneling if ingress filtering blocks that.

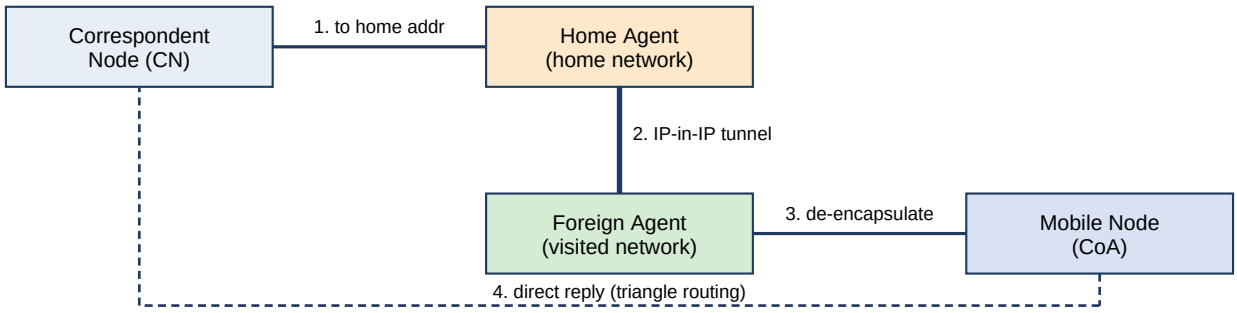

Figure 4: Mobile IP packet flow

Need: uninterrupted sessions (VoIP, downloads), a fixed identity for devices and servers, and transparency to upper layers and correspondent hosts. The drawback is triangle routing, which route optimization reduces.

Q10. Wired Ethernet frame vs wireless (802.11) frame

Ethernet II frame

```
| Dest MAC 6 | Src MAC 6 | Type 2 | Payload 46-1500 | FCS 4 |
```

IEEE 802.11 data frame

```
| Frame Ctrl 2 | Duration/ID 2 | Addr1 6 | Addr2 6 | Addr3 6 | Seq Ctrl 2 | [Addr4 6] | [QoS 2] | Payload 0-2304 | FCS 4 |
```

- Frame Control carries protocol version, type/subtype (management, control, data), To DS/From DS flags, Retry and Protected (encrypted) flags.
- Addr1 = receiver, Addr2 = transmitter, Addr3 = BSSID/source/destination depending on To DS/From DS. Addr4 is used only in AP-to-AP (WDS) frames.

Feature	Wired Ethernet (802.3)	Wireless LAN (802.11)
Wireshark filter	eth	wlan (needs monitor-mode capture)
Address fields	2 (Dest, Src)	3 (up to 4): Receiver, Transmitter, BSSID/DA/SA
Header size	14 B	24-30 B (+2 B QoS)
Frame types	Data only	Management (beacon, probe, auth, assoc), control (RTS, CTS, ACK), data
Type/Length field	Yes (EtherType)	No; type is in Frame Control, EtherType is in the LLC/SNAP header
Payload	46-1500 B	Up to 2304 B (MSDU)
Acknowledgement	None at layer 2	Per-frame ACK control frame
Retry/duplicates	None	Retry bit and sequence number
Encryption	None at layer 2	Optional (WPA2/3); Protected bit set
Medium access	CSMA/CD or full duplex	CSMA/CA with NAV (Duration)
Extra metadata	None	Radiotap header (signal, channel, rate) in captures

Documented Wireshark examples (typical values)

Ethernet: Dst: Cisco_ab:cd:ef (00:1a:2b:ab:cd:ef), Src: Dell_12:34:56 (00:14:22:12:34:56), Type: IPv4 (0x0800). The details pane shows 14 B of header, then IP; the 4 B FCS is usually stripped by the NIC.

802.11: Type/Subtype: Data (0x0020); Flags: 0x01 (To DS); Receiver (BSSID): AP MAC; Transmitter: Client MAC; Destination: Gateway MAC; Sequence: 1234; followed by a Radiotap header and LLC/SNAP (Type: IPv4). The payload is encrypted when the Protected flag is set.